

Вариант 7.

первая вершина: $A(0, 0, 0)$

Вершина B $(5, 0, 0)$

Вершина C $(0, 4, 2)$

Направление на свет $L(1, 0, 0)$

1. Векторы стороны

$$AB = B - A \rightarrow \begin{matrix} x & y & z \\ 5 & 0 & 0 \end{matrix}$$

$$AC = C - A \rightarrow \begin{matrix} x & y & z \\ 0 & 4 & 2 \end{matrix}$$

2) Векторное произв. $n = AB \times AC$

$$n_x = \underbrace{AB_y}_{0} \times \underbrace{AC_z}_{2} - \underbrace{AB_z}_{0} \times \underbrace{AC_y}_{4} = 0$$

$$n_y = \underbrace{AB_z}_{0} \times \underbrace{AC_x}_{0} - \underbrace{AB_x}_{5} \times \underbrace{AC_z}_{2} = -10$$

$$n_z = \underbrace{AB_x}_{5} \times \underbrace{AC_y}_{4} - \underbrace{AB_y}_{0} \times \underbrace{AC_x}_{0} = 20$$

$$3) |n| = \sqrt{\cancel{n_x^2} + 10^2 + 20^2} = \cancel{5\sqrt{5}} = 10\sqrt{5}$$

$$e = \left(\frac{0}{10\sqrt{5}}, \frac{-10}{10\sqrt{5}}, \frac{20}{10\sqrt{5}} \right) = \left(\frac{0}{\sqrt{5}}, \frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right) =$$

$$\left(0, \frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right) = (0, -0.447, 0.894)$$

$$n \cdot L = n_x \cdot L_x + n_y \cdot L_y + n_z \cdot L_z$$

$$0 \cdot 1 + (-10) \cdot 0 + 20 \cdot 0 = 0 = \cos \theta = \frac{n \cdot L}{|n| \cdot |L|} = \frac{0}{10\sqrt{5} \cdot 1} = 0$$